TASK SPECIFICATION: P4-03 PHASE P4 — FINANCIAL LEDGER & TUITION OFFSET ENGINE

# Tuition Account Management & Automatic Fee Offset API

Git Branch: `feature/P4-03-tuition` · Target Pull Request: feature/P4-03-tuition

**Core Objective & Technical Scope:**
Academic term tuition invoice accounts, automated credit offset from peer tutoring earnings, and unique fiscal receipt number generation.

| | | | |
|---|---|---|---|
| Priority Level | High | Complexity Rating | L |
| Execution Status | Backlog 📋 | Target PR / Commit | `feature/P4-03-tuition` |
| Upstream Dependencies | P4-02 | | |
| Architecture Blockers | None (Ready for execution) | | |

## IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Author V5__tuition.sql migration creating tuition_account and payment_receipt tables.
2. Implement TuitionService executing atomic tuition payment via wallet balance offset.
3. Implement sequential institutional receipt number generator.
4. Create TuitionController and DTOs with OpenAPI documentation.

## TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `backend/src/main/resources/db/migration/V5__tuition.sql`
- `wallet/application/Tuition*`
- `wallet/api/*`

## ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Offset amount cannot exceed available wallet balance or remaining invoice due.
- Receipt numbers are globally unique, sequential, and immutable.

## VERIFICATION & TEST COVERAGE STRATEGY

- TuitionServiceTest
- ReceiptGenerationTest
- TuitionControllerTest